fifty-fifty chance her qubit is spin-up or spin-down. Note the subtle difference: Alice's qubit is not in a quantum superposition where she doesn't know what the measurement outcome will be; it's in a state on each branch that will give a definite measurement outcome, but she doesn't know which state it is. We therefore describe her qubit as having a nonzero entropy. Von Neumann's idea was that we should ascribe a nonzero entropy to Alice's qubit even before Bob measures his, because after all she doesn't even know whether he's done a measurement. That's the entanglement entropy.

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Let's see how entanglement entropy appears in quantum field theory. Forgetting about gravity for a second, consider a region of empty space in the vacuum state, specified by a boundary separating inside the region from outside. Empty space is a richly textured place, full of quantum degrees of freedom that we can think of as modes of vibrating fields. The modes inside the region will be entangled with the modes outside, so the region has an entropy associated with it, even if the overall state is simply the vacuum.

We can even calculate what that entropy is. The answer is: infinity. This is a common complication with quantum field theory, that many questions of apparent physical relevance have seemingly infinite answers because there are an infinite number of possible ways for a field to vibrate. But just as we did for the vacuum energy in the last chapter, we can ask what happens when we impose a cutoff, allowing only modes longer than a certain wavelength. The resulting entropy is finite, and it turns out to be naturally proportional to the area of the region's boundary. The reason isn't hard to understand: field vibrations in one part of space are entangled with regions all over, but most of the entanglement is concentrated on nearby regions. The total entropy of a region of empty space depends on the amount of entanglement across the boundary, which is proportional to how big that boundary is—its area.

That's an intriguing feature of quantum field theory. Pick out a region within empty space, and the entropy of that region is proportional to the area of its boundary. That relates on the one hand a geometric quantity, the area of a region, to a "matter" quantity, the entropy contained inside. It all